Correcting the Weighted Hensel Estimate for Reed–Solomon Curve Decodability

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Abstract

The Reed–Solomon proximity-gap proof of Ben-Sasson, Carmon, Ishai, Kopparty and Saraf uses a weighted Hensel estimate to follow a root of an interpolation polynomial as X varies. We show that the estimate is false under its stated hypotheses, but also that the numerical bounds used later in the proof remain valid. For example, take

$$R(X,Y,Z) = Y + Z^2 + XY^2, \qquad x_0 = 0, \qquad H(Y,Z) = Y + Z^2.$$

The first Hensel numerator is then $\beta_1 = -Z^4$. Its weight is 4, while Claim A.2 gives the bound 1. The same proof later writes $\Lambda(\alpha_t) = \Lambda(Y) = 1$, although Λ has been defined only for polynomials and for regular quotient classes, not for arbitrary elements of the function field.

We repair the argument in two ways. The first keeps the original recurrence and records the degree allowance lost when the leading coefficient does not attain its permitted degree. The second clears denominators from the start and works entirely in the regular quotient $K[Z,T]/(\widehat{H})$. We carry both repairs through the removal of poles, truncation of the Hensel series, the common-numerator construction, the second resultant, and the final interpolation. For positive t, the two repaired numerators differ by the factor W^{t-1} and therefore give the same numerical bound.

A related issue occurs in the 2026 refinement: the degree of $R_i(x_0,Y,Z)$ does not control the coefficients obtained after shifting X. For instance, specializing $Y+Z+XZ^N$ at $X=0$ hides the term XZ^N. We replace the specialized degree by the full (Y,Z)-weighted degree of each global factor and prove that the improved threshold is preserved for both separable and inseparable factorizations.

All corrections and quantitative estimates needed for our conclusions are proved here. The direct correction starts from the published Hensel recurrence (A.1); the denominator-free proof derives its own recurrence and is independent of the disputed weight argument. The inverse-Frobenius construction follows the standard setup of the published Appendix C and is reproduced in the form needed below. A companion Lean 4 development independently checks the principal named results.

1 Introduction

Suppose a verifier is given several words $f_0, \dots, f_M$ and wants to test whether they are all close to a Reed–Solomon code. A standard trick is to choose a random field element z and test only the combined word

$$f_z = f_0 + zf_1 + \cdots + z^M f_M. \tag{1}$$

This saves queries and is used in FRI and related protocols [BBHR18, ACFY25, BGKTTZ23].

The saving creates a difficulty. For each value of z, a dishonest prover may point to a different nearby codeword. It is not enough to know that many of the words f_z are individually close to the code. We must show that many of the chosen codewords belong to one polynomial family,

$$c_z = C_0 + zC_1 + \cdots + z^M C_M. \tag{2}$$

Once this is known, a root count in the variable z gives a common agreement set for the original words. This conclusion is called curve decodability.

Ben-Sasson, Carmon, Ishai, Kopparty and Saraf prove curve decodability by working over the rational function field $K(Z)$ [BCIKS23]. Their argument begins with Guruswami–Sudan interpolation [GS99]. It then chooses an irreducible factor $R(X, Y, Z)$ of the interpolation polynomial and a factor $H(Y, Z)$ of $R(x_0, Y, Z)$ for a suitable value x_0. Hensel lifting follows one root of H as X moves away from x_0.

Write the lifted root as

$$\alpha_0 + \alpha_1(X - x_0) + \alpha_2(X - x_0)^2 + \cdots .$$

The published proof needs a weight bound on these coefficients three times. First it removes challenge values at which a denominator vanishes. Next it shows that all coefficients above the desired X-degree are zero. Finally it puts the remaining coefficients over one denominator and proves that the truncated root has the required degree in Z. Claim A.2 supplies the estimate used at each of these steps.

The estimate fails when the leading coefficient of the chosen branch has smaller degree than the maximum allowed by the hypotheses. At order zero, the proof identifies the weight of T with $\Lambda(W) + 1$, although it has only proved the corresponding inequality. Later it writes $\Lambda(\alpha_t) = \Lambda(Y) = 1$, even though the weight is defined only for polynomials and for regular quotient classes. In the example displayed in the abstract, the first positive numerator has weight 4 while the printed bound is 1. Section 3 gives the exact source statements and checks the examples against all algebraic hypotheses.

There are two useful repairs. Section 4 keeps the original recurrence and proves the missing weight induction. Sections 6 and 7 give a second proof after clearing denominators. This form stays inside the regular quotient and is especially well suited to formal verification. Both proofs lead to the same numerical estimate and both support the two resultant arguments needed for curve decodability.

The 2026 refinement introduces a separate degree question. A bound on $R_i(x_0, Y, Z)$ may hide large coefficients multiplied by $X - x_0$. We show that the full (Y, Z)-weighted degree of the global factor R_i gives the correct bound after shifting X, and that these full degrees still fit the refined global count. The argument is given for both separable factors and factors of the form $R_i(X, Y^{p^f}, Z)$.

Anyone formalizing the smooth-branch proof from the published definitions meets the original problem immediately: α_t belongs to the function field, while Λ is defined on the regular quotient. The correction therefore requires an actual induction on regular representatives. We give that induction in the original notation so that it can be reused directly.

The direct repair starts from the published recurrence (A.1), whose weight analysis is the object being corrected. The denominator-free proof derives an independent recurrence from the shifted coefficients of R. All corrections and quantitative estimates used later are proved in this paper. The standard inverse-Frobenius setup from Appendix C is reproduced in the precise form needed here. A separate Lean 4 development, described in Section 8, checks the principal named results independently.

2 Curve decodability

Let K be a finite field, and let $D = (x_1, \ldots, x_n)$ be a list of distinct elements of K. The Reed–Solomon code of dimension k consists of the evaluation vectors of polynomials of degree below k:

$$\mathrm{RS}_K(D, k) = \{(p(x_1), \ldots, p(x_n)) : p \in K[X],\ \deg p < k\}.$$

A generalized Reed–Solomon code multiplies the ith coordinate by a fixed nonzero scalar v_i. Dividing that coordinate by v_i returns the ordinary Reed–Solomon normalization and does not change the set of agreeing coordinates.

Fix words $f_0, \ldots, f_M \in K^D$ and a challenge set $S \subseteq K$. For each $z \in S$, form the combined word f_z from (1). An adversary may answer with a codeword c_z and an agreement set $A_z \subseteq D$. Both choices may depend on z. For a support threshold θ, define

$$G = \{z \in S : |A_z| > \theta \text{ and } f_z(x) = c_z(x) \text{ for every } x \in A_z\}.$$

Thus G is the set of challenges for which the claimed nearby codeword really does agree on more than θ coordinates.

The next definition states the conclusion sought from the algebraic argument. It says that sufficiently many successful challenges force many of the codewords c_z to lie on one degree-M curve.

Definition 2.1 (Curve decodability)**.** A linear code $C \subseteq K^D$ is (M, θ, a, b)-curve-decodable over S if $|G| > a$ implies the existence of $C_0, \ldots, C_M \in C$ and $G' \subseteq G$ such that $|G'| > b$ and

$$c_z = \sum_{j=0}^{M} z^j C_j \qquad (z \in G').$$

The lower bound on $|G'|$ has a simple purpose. Suppose that, at some coordinate x, at least one equality $f_j(x) = C_j(x)$ fails. Then

$$\sum_{j=0}^{M} z^j (f_j(x) - C_j(x))$$

is a nonzero polynomial in z of degree at most M. It vanishes for at most M challenge values. Taking the union over all coordinates shows that at most $M|D|$ challenges can conceal a discrepancy somewhere. The correlated-agreement application therefore chooses $b > M|D|$.

In the complete proximity-gap proof, multiplicity interpolation and two factor selections reduce the analysis to one fixed pair (R, H). The present paper repairs the weight estimate used for that pair. Once the repair is in place, the root counts and the final interpolation produce the curve of codewords in the definition above.

3 Where the estimate in Claim A.2 fails

We first use the notation of the journal paper. This lets the disputed equalities be checked directly against Claim A.2, before we introduce the slightly more general notation used in the repair. The chosen branch is written as

$$H(Y, Z) = h_0(Z)Y^h + h_1(Z)Y^{h-1} + \cdots + h_h(Z), \qquad W = h_0, \tag{3}$$

with

$$\deg_Z h_k \leq D + k - h. \tag{4}$$

The leading coefficient W need not have the largest degree allowed by this inequality.

The change of variables $T = WY$ turns the nonmonic equation $H(Y, Z) = 0$ into a monic equation in T. Appendix A.2 assigns weight 1 to Z and weight

$$\Lambda(T) = D + 1 - h \tag{5}$$

to T. Thus $Z^a T^b$ has weight $a + b(D + 1 - h)$. For polynomials, the weight is the largest weight of a nonzero monomial. For residue classes, the paper first reduces to the canonical representative of degree below h in T. The weight is additive before reduction and only subadditive afterwards [BCIKS23, JACM pp. 31:46–31:47].

For the correction we allow different degree bounds for the parent polynomial and the chosen branch. We also index the branch by increasing powers of Y:

$$H(Y, Z) = \sum_{i=0}^{h} a_i(Z) Y^i, \qquad W = a_h \neq 0. \tag{6}$$

Assume

$$\deg a_i + \ell i \leq D_H \qquad (a_i \neq 0). \tag{7}$$

Here Y has weight ℓ and Z has weight 1. Set

$$b_H = D_H - \ell h, \qquad \tau = b_H + \ell = D_H + \ell - \ell h. \tag{8}$$

The number b_H is the largest degree permitted for the leading coefficient W. If $w = \deg W$ is its actual degree, then

$$w + \ell \leq \tau. \tag{9}$$

The printed proof uses equality here. The difference $b_H - w$ is the unused degree allowance that must be retained.

The monic equation is

$$\widehat{H}(T, Z) = W^{h-1} H(T/W, Z) = T^h + \sum_{i=0}^{h-1} a_i W^{h-1-i} T^i. \tag{10}$$

We use two related rings:

$$L = K(Z)[Y]/(H), \qquad \mathcal{O} = K[Z, T]/(\widehat{H}). \tag{11}$$

Write $y \in L$ for the residue class of Y. The substitution $T = WY$ canonically identifies $K(Z)[T]/(\widehat{H})$ with L. The smaller ring $\mathcal{O}$ contains the classes that have polynomial representatives in $K[Z, T]$. These are the regular elements, and they carry the weight above. A general element of L may contain a denominator in Z and need not have a weight.

The choice of τ ensures that division by the monic relation does not raise the weight. Indeed, each lower coefficient of $\widehat{H}$ satisfies

$$\deg(a_i W^{h-1-i}) + i\tau \leq h\tau. \tag{12}$$

The identity

$$D_H + (h-1) b_H = h(b_H + \ell)$$

explains this bound. It uses the permitted degree b_H, not the actual degree w.

3.1 What the printed proof establishes

Claim A.2 writes the Hensel coefficients as

$$\alpha_t = \frac{\beta_t}{W^{t+1}\xi^{e_t}}, \qquad \beta_t \in \mathcal{O}, \qquad e_0 = 0, \quad e_t = 2t-1 \ (t > 0). \tag{13}$$

Its proof performs two different tasks, and it is important not to confuse them.

The first task is to prove that every numerator β_t is regular. Recurrence (A.1) expresses β_t using earlier numerators and nonnegative powers of W and ξ. This part of the argument is correct. There is one harmless slip in the surrounding prose. When the shifted X-order equals t, the relevant partition is empty, so its size is zero rather than at least one. The exponent of ξ is then $2t-2$, which is still nonnegative.

The second task is to bound the weight of the numerator. Before stating the recurrence, the paper writes

$$\Lambda(B_{s,\lambda}) = (D-r) + (d-r-\varepsilon_s)\Lambda(W), \tag{14}$$

where $r = |\lambda|$, while ε_s is 1 for $s = 0$ and 0 otherwise [BCIKS23, JACM p. 31:49]. This cannot be an exact weight in general: coefficients may vanish, terms may cancel, and reduction modulo $\widehat{H}$ may lower the weight. At most, it can be an upper bound.

At order zero, $\beta_0 = T$. The proof then says "$\Lambda(T) = \Lambda(W) + 1$" [BCIKS23, JACM Claim A.2, p. 31:48]. Later it does not carry out the suggested induction on recurrence (A.1). Instead it writes

$$\Lambda(\alpha_t) = \Lambda(Y) = 1 \tag{15}$$

and concludes

$$\Lambda(\beta_t) \le 1 + (t+1)\Lambda(W) + e_t\Lambda(\xi). \tag{16}$$

This passage appears on p. 31:50 of the journal version [BCIKS23].

The problem is not subtle. The weight Λ has been defined on $K[T,Z]$ and on regular classes in $\mathcal{O}$. It has not been defined on arbitrary elements of L. Neither α_t nor $Y = T/W$ need be regular. If one reads $\Lambda(Y) = 1$ as $\Lambda(T) - \Lambda(W) = 1$, this assumes the failed order-zero equality. The final step to (16) also uses additive weight arithmetic with α_t, an element outside the rings on which those laws were proved.

Equation (15) is a useful mnemonic, but it is not a proof under the definitions given in the paper. There is no weight induction whose constant can be adjusted. A repair must supply the missing induction on regular representatives. We do so in Section 4.

For completeness, the statement changed once before publication. The July 2020 manuscript used equality in its first displayed estimate for β_t. The 2021 revision changed that relation to an inequality and noted that the quotient weight is subadditive. It retained both the order-zero equality and (15). The 2023 journal version agrees with the 2021 revision at these passages [BCIKS20, BCIKS21, BCIKS23].

3.2 Examples where the stated bounds fail

The first example isolates the missing saturation assumption. In the source notation, take $h = 1$, $D = 2$, $h_0 = 1$, and $h_1 = Z^2$. Then $H(Y,Z) = Y + Z^2$ satisfies every bound in (4), but

$$\Lambda(T) = 2 > 1 = \deg W + 1. \tag{17}$$

Even if D is the exact weighted degree of H, the maximum need not occur at the leading coefficient. Here it occurs at the constant coefficient.

This example concerns only the branch polynomial. The next one includes the parent polynomial and satisfies the irreducibility and separability conditions used by the Hensel argument. It also shows that the failure reaches the first positive coefficient, not only the base case.

Proposition 3.1 (A sharp coefficient counterexample)**.** *Let K be any field and set*

$$R(X,Y,Z) = Y + Z^2 + XY^2, \qquad x_0 = 0, \qquad H(Y,Z) = Y + Z^2. \tag{18}$$

Then R is irreducible and separable in Y, while $H = R(0,Y,Z)$ is a simple irreducible branch. With $\ell = 1$ one has

$$D_R = D_H = 2, \qquad d = 2, \qquad h = 1, \qquad W = 1,$$

so $b_H = 1$, $w = 0$, and $\tau = 2$. The source derivative numerator is $\xi = 1$. Write the Hensel root as

$$\alpha(U) = \sum_{t \geq 0} \alpha_t U^t.$$

Then

$$\alpha_0 = -Z^2, \qquad \beta_1 = \alpha_1 = -Z^4, \qquad \Lambda(\beta_1) = 4. \tag{19}$$

The printed bound (16) *gives* 1*. Replacing only its initial* 1 *by τ gives* 2*, which is still false. The corrected ceiling in Theorem 4.4 is $P_1 = 4$, so the replacement estimate is attained in this example.*

Proof. View R as the primitive linear polynomial $Y^2X + (Y + Z^2)$ in $K[Y,Z][X]$. Its two coefficients are coprime, so Gauss's lemma gives irreducibility. Its Y-derivative is $1 + 2XY$, which is nonzero; since R is irreducible, it is separable in Y. At $X = 0$, the selected branch is linear and has derivative 1.

The Hensel equation is

$$\alpha(U) + Z^2 + U\alpha(U)^2 = 0.$$

The coefficients of U^0 and U^1 give $\alpha_0 = -Z^2$ and $\alpha_1 = -\alpha_0^2 = -Z^4$. Since $W = \xi = 1$, the source numerator is $\beta_1 = \alpha_1$. Its weight is its Z-degree, which is 4.

For the term of (A.1) with shifted-X order 1 and empty partition, the source auxiliary numerator is T^2. In the quotient by $\widehat{H} = T + Z^2$, its canonical representative is Z^4, again of weight 4. The printed upper bound gives $D_R = 2$. □

The discrepancy is not confined to the choice $D_R = 2$. For every integer $D \geq 2$, take

$$R_D(X,Y,Z) = Y + Z^D + XY^2, \qquad H_D(Y,Z) = Y + Z^D.$$

The same argument gives $\beta_1 = -Z^{2D}$, of weight $2D$, while the printed bound remains 1. Even replacing that initial 1 by the permitted generator weight $\tau = D$ still misses by D. Thus the estimate cannot be repaired by changing only its base constant.

Remark 3.2 (The derivative-numerator bound can also fail). Over $K = \mathbb{F}_5$, let

$$R(X,Y,Z) = (Y + Z^2)(Y^2 + 1) + X, \qquad H(Y,Z) = Y + Z^2, \qquad x_0 = 0.$$

Then R is irreducible and separable in Y, $R(0,Y,Z)$ is square-free, and the selected branch is simple. Here $D_R = 4$, $D_H = 2$, $d = 3$, $h = 1$, and $W = 1$. At the root $Y = -Z^2$,

$$\xi = \partial_Y R(0, -Z^2, Z) = Z^4 + 1,$$

so $\Lambda(\xi) = 4$. The first displayed upper bound for ξ in Claim A.2 is 3. The corrected bound in Corollary 4.3 below is 5 when the separate parent and branch degrees are retained.

The estimates in Claim A.2 are therefore false under the stated hypotheses. This does not make the curve-decodability theorem false. The same recurrence and the same algebraic root can still be used, provided the degree slack is counted correctly. We now give that correction.

4 Repairing the original recurrence

We first repair the published recurrence in place. The notation is longer than the idea. When a power of the Hensel series is expanded, we must record how many copies of each earlier coefficient occur and how their orders add up. That is all the partition notation will do.

Let $R \in K[X, Y, Z]$ have Y-degree d. Assume that every monomial $X^a Y^j Z^b$ occurring in R satisfies

$$b + \ell j \leq D_R, \tag{20}$$

and assume explicitly that

$$H(Y, Z) \mid R(x_0, Y, Z) \qquad \text{in } K[Z, Y]. \tag{21}$$

The second hypothesis is literal divisibility in the polynomial ring $K[Z, Y]$. This is stronger and clearer than saying only that H is a factor after passing to $K(Z)[Y]$. It is the form supplied by the factor selection in the proximity-gap proof. One may equivalently begin with primitive factors and use Gauss's lemma.

Lemma 4.1 (Translation preserves the parent degree bound)**.** *Write*

$$R(x_0 + U, Y, Z) = \sum_{s \geq 0} \sum_{j=0}^{d} c_{s,j}(Z) U^s Y^j. \tag{22}$$

Then every nonzero coefficient satisfies

$$\deg c_{s,j} + \ell j \leq D_R. \tag{23}$$

Proof. Give X and U weight zero, Y weight ℓ, and Z weight one. Replacing X by $x_0 + U$ changes only the coefficient of a monomial; it does not change its powers of Y or Z. Every monomial contributing to $c_{s,j}(Z)U^s Y^j$ therefore had weight at most D_R before the translation. Cancellation can only lower the resulting Z-degree. □

We now unpack one coefficient of the recurrence. Following JACM pp. 31:48–31:50 [BCIKS23], let $\lambda = (\lambda_u)_{u \geq 1}$ record how many copies of each earlier coefficient α_u occur. Its total U-degree is $n = \sum_{u \geq 1} u\lambda_u$. Put

$$r = |\lambda| = \sum_{u \geq 1} \lambda_u. \tag{24}$$

Thus r counts factors, while n counts total order. At shifted X-order s, the coefficient of $\prod_{u \geq 1} \alpha_u^{\lambda_u}$ is

$$A_{s,\lambda} = \sum_{j=r}^{d} \kappa_{j,\lambda} c_{s,j}(Z) \alpha_0^{j-r}, \tag{25}$$

where $\kappa_{j,\lambda} \in K$ is the relevant multinomial coefficient. If this scalar is zero, the term disappears. We therefore discuss only nonzero $A_{s,\lambda}$, for which necessarily $r \leq d$.

Let ε_s be 1 when $s = 0$ and 0 otherwise. Define

$$N_{s,\lambda} = W^{d-r} A_{s,\lambda}. \tag{26}$$

Using $\alpha_0 = T/W$, this element has the polynomial representative

$$N_{s,\lambda} = \sum_{j=r}^{d} \kappa_{j,\lambda} c_{s,j}(Z) T^{j-r} W^{d-j}. \tag{27}$$

When $s = 0$, the numerator contains one additional factor of W. For $j < d$ this factor is visible in (27). For $j = d$, it follows from $H \mid R(x_0, Y, Z)$: the leading coefficient W of H divides the leading Y-coefficient of the parent. We may therefore choose $B_{s,\lambda} \in \mathcal{O}$ so that

$$N_{s,\lambda} = W^{\varepsilon_s} B_{s,\lambda}. \tag{28}$$

Set

$$w = \deg W, \qquad \mu = (D_R - \ell) + (d-1)b_H, \qquad \sigma = \mu - w. \tag{29}$$

In the selected-factor setting,

$$D_R \geq D_H, \qquad d \geq h \geq 1, \qquad w \leq b_H.$$

Moreover, $D_H - \ell \geq b_H \geq w$. Hence $D_R - \ell \geq w$, so $\mu \geq w$ and $\sigma \geq 0$.

Lemma 4.2 (Weight of the auxiliary numerator $B_{s,\lambda}$)**.** *Assume* (21)*. For every pair* (s, λ) *with* $A_{s,\lambda} \neq 0$*, the element chosen in* (28) *satisfies*

$$\Lambda(B_{s,\lambda}) \leq D_R - \ell r + (d-r)b_H - \varepsilon_s w. \tag{30}$$

This is only an upper bound. Vanishing coefficients, cancellation, or reduction modulo $\widehat{H}$ *may lower the actual weight. Compared with the analogue of* (14)*, the missing allowance is*

$$(d-r)(b_H - w). \tag{31}$$

Proof. A summand of (27) has weight at most

$$D_R - \ell r + (j-r)b_H + (d-j)w. \tag{32}$$

Because $w \leq b_H$, the quantity in (32) is at most $D_R - \ell r + (d-r)b_H$. The whole sum has the same upper bound.

If $s > 0$, nothing more is needed. If $s = 0$, remove the additional factor W identified above. Multiplication by the nonzero polynomial W adds exactly w to the weight of a polynomial representative. Dividing out this known factor therefore subtracts w. Passing to the quotient can only lower the weight further. □

For $s = 0$ and $r = 1$, the element $B_{0,\lambda}$ is the source derivative numerator ξ. Equivalently, $W^{d-1}\zeta = W\xi$ in the function field, which is the source notation $\xi = W^{d-2}\zeta$.

Corollary 4.3 (Weight of the derivative numerator ξ)**.** *The source numerator* ξ *satisfies*

$$\Lambda(\xi) \leq \sigma. \tag{33}$$

Proof. Apply Lemma 4.2 with $s = 0$ and $r = 1$. Its right side is

$$(D_R - \ell) + (d-1)b_H - w = \sigma.$$

□

Let $\lambda^{(t)}$ denote the partition with one part t. The recurrence (A.1) on JACM p. 31:50, in the present notation, is

$$\beta_t = \sum_{\substack{0 \leq s \leq t,\ \lambda \vdash (t-s) \\ (s,\lambda) \neq (0,\lambda^{(t)})}} W^{s+\varepsilon_s-1} \xi^{2s+r-2} B_{s,\lambda} \prod_{u \geq 1} \beta_u^{\lambda_u}. \tag{34}$$

The excluded term is the unique occurrence of α_t moved to the left-hand derivative factor. Equation (34) is quoted from the published proof; the result below repairs its weight analysis. The later reconstruction does not depend only on this quoted recurrence: Section 6 derives a denominator-free recurrence directly from the shifted coefficients of R, and Section 7 uses that independent construction.

Theorem 4.4 (Corrected bound for the original Hensel numerators)**.** *Let $\beta_0 = T$ and let the remaining β_t satisfy* (34)*. Put*

$$P_t = \tau + tw + e_t\sigma. \tag{35}$$

Then

$$\Lambda(\beta_t) \le P_t \qquad (t \ge 0). \tag{36}$$

For $t > 0$, the same upper bound can be written as

$$P_t = \tau + e_t\mu - (t-1)w. \tag{37}$$

Proof. The base case. The base case is $\beta_0 = T$, so $\Lambda(\beta_0) \le \tau = P_0$. No equality between τ and $w + \ell$ is used.

The induction step. Fix $t > 0$ and assume the estimate for all smaller indices. In one term of (34), put

$$n = \sum_{u \ge 1} u\lambda_u = t - s, \qquad r = \sum_{u\ge 1} \lambda_u.$$

Before estimating the weight, check that the term is well defined. If $s = 0$, removing the singleton term leaves at least two positive parts. If $0 < s < t$, then $r \ge 1$. If $s = t$, the partition is empty and the exponent of ξ is $2t - 2$. In every case the exponents of W and ξ are nonnegative, and every index u with $\lambda_u > 0$ is smaller than t.

The product of earlier coefficients. Since $e_u = 2u - 1$ for positive u,

$$\sum_{u\ge 1} \lambda_u e_u = 2n - r. \tag{38}$$

The induction hypothesis therefore gives

$$\Lambda\left(\prod_{u\ge 1} \beta_u^{\lambda_u}\right) \le r\tau + nw + (2n - r)\sigma. \tag{39}$$

One term of the recurrence. Let $\mathcal{T}_{s,\lambda}$ be the corresponding summand in (34). Applying Lemma 4.2 and Corollary 4.3 and (39) gives

$$\begin{aligned}\Lambda(\mathcal{T}_{s,\lambda}) \le{}& (s + \varepsilon_s - 1)w + (2s + r - 2)\sigma \\ &+ D_R - \ell r + (d - r)b_H - \varepsilon_s w \\ &+ r\tau + nw + (2n - r)\sigma.\end{aligned} \tag{40}$$

We now collect like terms. The w-terms give

$$(s + \varepsilon_s - 1) - \varepsilon_s + n = t - 1. \tag{41}$$

The σ-terms give

$$(2s + r - 2) + (2n - r) = 2t - 2 = e_t - 1. \tag{42}$$

The remaining terms give

$$D_R - \ell r + (d - r)b_H + r\tau = D_R + db_H. \tag{43}$$

Finally,

$$\tau + w + \sigma = D_R + db_H. \tag{44}$$

Finishing the induction. Substitution of (44) into (40) gives $\Lambda(\mathcal{T}_{s,\lambda}) \le \tau + tw + e_t\sigma = P_t$. Every summand has this bound, and the weight of a sum is at most the largest weight of its summands. This proves (36). Equation (37) follows from $\sigma = \mu - w$ and $e_t = 2t - 1$. $\square$

The cancellation is exact at the level of these upper bounds: no term depending on s or λ remains. The actual weight can still be smaller, since coefficients may vanish, terms may cancel, and quotient reduction may lower the weight.

It is also useful to compare the answer with the expression printed in Claim A.2. If the leading coefficient attains its permitted degree, so that $w = b_H$, then $\tau = w + \ell$ and

$$\sigma = (D_R - \ell) + (d-2)w.$$

Thus P_t has the form obtained from the printed estimate after replacing $\Lambda(\xi)$ by its corrected upper bound. It need not equal the expression using the actual value of $\Lambda(\xi)$, since ξ may not attain that upper bound.

4.1 Why the rest of the counting argument still works

A corrected estimate is useful only if it is still small enough for the later root counts. The published proof spends its allowance in three places, and we check them one at a time.

Let D_* be a common upper bound for D_R and D_H. Assume $1 \le h \le d$ and $\ell \ge 1$. Then

$$\tau \le D_*, \qquad \mu \le dD_* - \ell < dD_*. \tag{45}$$

First, the proof removes the values of z at which W or ξ vanishes. Their total number is at most

$$w + h\Lambda(\xi) \le w + h\sigma = h\mu - (h-1)w < hdD_*. \tag{46}$$

Second, one resultant is used for each coefficient above the desired degree. For every $t > 0$,

$$hP_t \le h(\tau + e_t\mu) < h(2t+1)dD_*. \tag{47}$$

Indeed, $P_t \le \tau + e_t\mu \le D_* + (2t-1)dD_* \le 2tdD_*$. This is exactly the coarse allowance used to force the high coefficients to vanish.

Third, after the high coefficients have vanished, the remaining coefficients are put over one common denominator:

$$\beta_m^{\mathrm{com}}(X) = \sum_{t=0}^{m} \beta_t (X - x_0)^t W^{m-t} \xi^{e_m - e_t}. \tag{48}$$

Each summand has weight at most P_m. Indeed, its upper bound is $P_t + (m-t)w + (e_m - e_t)\sigma = P_m$. If $w_x(Z)$ has degree at most ℓ, then the target term satisfies

$$\Lambda\big(w_x W^{m+1} \xi^{e_m}\big) \le P_m. \tag{49}$$

The only input is $\deg w_x + w \le \tau$. Therefore the difference between the common numerator and the proposed polynomial in Z also has weight at most P_m. Equation (47), now with $t = m$, pays for the second resultant.

Thus the corrected induction fits all three later uses: pole removal, truncation of the series, and recovery of the Z-dependence.

5 How the fixed branch arises

The corrected Hensel estimate is local: it concerns one parent factor R and one branch factor H. We briefly recall how the proximity-gap proof chooses this pair.

Let G be the set of challenges for which the prover supplies a nearby codeword. Multiplicity interpolation gives a nonzero polynomial

$$Q(X,Y,Z) \in K[X,Y,Z]$$

of controlled degree such that every selected polynomial $p_z(X)$ satisfies

$$Q(X,p_z(X),z) = 0 \qquad (z \in G). \tag{50}$$

Factor Q into irreducibles. Each challenge must be accounted for by at least one factor. Counting the challenges assigned to the factors therefore selects one factor $R(X,Y,Z)$ that accounts for a large subset of G.

The proof next chooses $x_0 \in K$ away from a separability resultant. This ensures that $R(x_0,Y,Z)$ is square-free over $K(Z)$. Factor the specialized polynomial. A second count selects an irreducible factor $H(Y,Z)$ and a set $G_{R,H}$ for which

$$H(p_z(x_0),z) = 0, \qquad R(X,p_z(X),z) = 0 \qquad (z \in G_{R,H}). \tag{51}$$

Only existence is used. The proof does not require an algorithm that finds the factors.

Recall that $L = K(Z)[Y]/(H)$, and write y for the residue class of Y. The choice of x_0 makes y a simple root of the parent equation:

$$R(x_0,y,Z) = 0, \qquad \zeta := \partial_Y R(x_0,y,Z) \neq 0 \quad \text{in } L. \tag{52}$$

Hensel's lemma therefore gives a unique series

$$\alpha(U) = \sum_{t\geq 0} \alpha_t U^t, \qquad \alpha_0 = y, \qquad R(x_0+U,\alpha(U),Z) = 0. \tag{53}$$

By this stage, the published proof has already removed challenge values at which the interpolation polynomial vanishes identically, two factors collide, the chosen root is not simple, or $W(z) = 0$. What remains is a numerical question: how many additional challenges can be lost because of the Hensel coefficients? The next two sections answer it.

6 A denominator-free proof in the regular quotient

6.1 The ring of regular elements

The purpose of $T = WY$ is to keep polynomial representatives free of denominators. The regular classes form $\mathcal{O} = K[Z,T]/(\widehat{H})$. Since $\widehat{H}$ is monic of degree h in T, ordinary polynomial division gives every class a unique representative of T-degree below h. We measure the weight of that representative.

There is also a natural map from the regular quotient to the branch field:

$$\iota : \mathcal{O} \longrightarrow L, \qquad T \longmapsto Wy, \tag{54}$$

because $\widehat{H}(Wy,Z) = W^{h-1}H(y,Z) = 0$.

Lemma 6.1 (Embedding the regular quotient in the function field)**.** *The map ι is injective. Consequently $\mathcal{O}$ is an integral domain. Moreover, $\widehat{H}$ is irreducible over $K(Z)$.*

Proof. Let $u \in \mathcal{O}$ have canonical representative $P(T,Z) = \sum_{i=0}^{h-1} p_i(Z)T^i$. If $\iota(u) = 0$, then $P(Wy,Z) = 0$ in L. Hence $H(Y,Z)$ divides $P(WY,Z)$ in $K(Z)[Y]$. The latter polynomial has Y-degree less than $h = \deg_Y H$, so it is zero. Since $W \neq 0$, every coefficient $p_i W^i$ is zero in $K(Z)$, and therefore $p_i = 0$ in $K[Z]$. Thus $u = 0$.

Over $K(Z)$, the change of variables $T = WY$ is invertible. The identity

$$\widehat{H}(WY, Z) = W^{h-1} H(Y, Z)$$

then shows that irreducibility of H is preserved by this change of variables and multiplication by the nonzero scalar W^{h-1}. □

For each i, let $p_i(Z)$ denote the coefficient of T^i in a polynomial $P(T, Z)$. Its weight is

$$\Lambda(P) = \max_{p_i \neq 0} \{\deg p_i + i\tau\}, \qquad \Lambda(0) = -\infty. \tag{55}$$

For a class in $\mathcal{O}$, we take the weight of its unique representative of T-degree below h.

Lemma 6.2 (Reduction does not increase the weight)**.** *For every $P \in K[Z, T]$,*

$$\Lambda(P \bmod \widehat{H}) \leq \Lambda(P).$$

Proof. Write $\widehat{H} = T^h + \sum_{i<h} b_i(Z) T^i$. Equation (12) gives $\deg b_i + i\tau \leq h\tau$. A single division step replaces $c(Z)T^{h+r}$ by the lower powers $-c(Z)b_i(Z)T^{i+r}$. Their weights obey

$$\deg(cb_i) + (i + r)\tau \leq \deg c + (h - i)\tau + (i + r)\tau = \deg c + (h + r)\tau.$$

Each step lowers the largest power of T, and the displayed inequality shows that it does not raise the weight. Repeating the step proves the lemma. □

Corollary 6.3 (Basic weight inequalities)**.** *For $u, v \in \mathcal{O}$,*

$$\Lambda(u + v) \leq \max\{\Lambda(u), \Lambda(v)\},$$
$$\Lambda(uv) \leq \Lambda(u) + \Lambda(v).$$

For every integer $r \geq 1$,

$$\Lambda(u^r) \leq r\Lambda(u). \tag{56}$$

The zeroth power is handled separately by $u^0 = 1$ and $\Lambda(1) = 0$.

Proof. Addition is coefficientwise. For multiplication, multiply canonical representatives and apply Lemma 6.2. Equation (56) follows by induction on r. □

6.2 Clearing the shifted equation

We next clear the denominators created by substituting $Y = T/W$ into the shifted equation. In the term $c_{s,j} Y^j$, the factor W^j appears in the denominator. Multiply by W^d and set

$$\rho_{s,j} = c_{s,j} W^{d-j} \in K[Z] \subseteq \mathcal{O}. \tag{57}$$

The same operation applied to the derivative gives the regular element

$$\eta := \sum_{j=1}^{d} j\, c_{0,j} W^{d-j} T^{j-1} \in \mathcal{O}. \tag{58}$$

Under (54),

$$\iota(\eta) = W^{d-1}\zeta. \tag{59}$$

The quantities μ and σ are those of (29). In source notation $\eta = W\xi$.

We shall later cancel powers of W and η, so we record their nonvanishing now. The polynomial W is nonzero, hence nonzero in $K(Z)$. Smoothness gives $\zeta \neq 0$ in L, and (59) then gives $\iota(\eta) \neq 0$. Since ι is injective, $\eta \neq 0$ in $\mathcal{O}$. Thus $W\eta^{e_t}$ is nonzero for every t.

Lemma 6.4 (Weights of the cleared derivative and coefficients)**.** *The elements in* (57) *and* (58) *satisfy*

$$\Lambda(\eta) \leq \mu, \tag{60}$$

$$\Lambda(\rho_{s,j}) + j\tau \leq \tau + \mu. \tag{61}$$

Proof. For a term of (58), the definition of τ gives the upper bound

$$D_R - \ell j + (d-j)b_H + (j-1)(b_H + \ell).$$

The right-hand side simplifies to μ. The same calculation for $\rho_{s,j}$ gives

$$\Lambda(\rho_{s,j}) + j\tau \leq D_R - \ell j + (d-j)b_H + j(b_H + \ell).$$

The expression on the right is $D_R + db_H = \tau + \mu$. □

6.3 A recurrence with no division

We now build the cleared coefficient of order t without first forming a fraction in L. A tuple $\mathbf{u} = (u_1, \ldots, u_j)$ records the coefficient chosen from each of j copies of the Hensel series. For $t > 0$, let

$$\mathcal{P}_{s,j,t} \coloneqq \{(u_1, \ldots, u_j) \in \mathbb{N}^j : s + u_1 + \cdots + u_j = t\}. \tag{62}$$

When $s = 0$, a tuple with exactly one positive entry contributes the new coefficient α_t. These are precisely the linear terms that form the derivative factor. Move them to the left and remove them from the sum. Call the remaining set $\mathcal{P}^\circ_{s,j,t}$. For $s > 0$, no term is removed, so $\mathcal{P}^\circ_{s,j,t} = \mathcal{P}_{s,j,t}$.

The coefficient of U^t in the Hensel equation is then

$$\zeta\alpha_t = -N_t, \qquad N_t = \sum_{s=0}^{t}\sum_{j=0}^{d} c_{s,j} \sum_{\mathbf{u}\in\mathcal{P}^\circ_{s,j,t}} \prod_{i=1}^{j} \alpha_{u_i}. \tag{63}$$

The sum N_t contains only coefficients of order below t. The removed linear terms are exactly $\zeta\alpha_t$.

Define $e_0 = 0$ and $e_t = 2t - 1$ for $t > 0$. For a tuple $\mathbf{u} = (u_1, \ldots, u_j)$, let $r(\mathbf{u})$ be the number of positive entries. Then

$$\sum_{i=1}^{j} e_{u_i} = 2\sum_{i=1}^{j} u_i - r(\mathbf{u}). \tag{64}$$

If $s = 0$, every retained tuple has at least two positive entries, so the right-hand side is at most $2t - 2 = e_t - 1$. If $s > 0$, then $\sum_i u_i = t - s \leq t - 1$, which gives the same bound. Hence

$$\omega(\mathbf{u}) = e_t - 1 - \sum_{i=1}^{j} e_{u_i} \tag{65}$$

is nonnegative. It supplies exactly the extra power of η needed to give every summand the same clearing factor. Every retained index u_i is smaller than t, so the recurrence is genuinely inductive.

Theorem 6.5 (Cleared Hensel coefficients)**.** *There are elements $\delta_t \in \mathcal{O}$ satisfying*

$$\iota(\delta_t) = W\iota(\eta)^{e_t}\alpha_t \tag{66}$$

and

$$\Lambda(\delta_t) \le \tau + e_t\mu \qquad (t \ge 0). \tag{67}$$

They are given recursively by

$$\delta_0 = T \tag{68}$$

and, for $t > 0$,

$$\delta_t := -\sum_{s=0}^{t}\sum_{j=0}^{d}\sum_{\mathbf{u}\in\mathcal{P}^{\circ}_{s,j,t}} \rho_{s,j}\eta^{\omega(\mathbf{u})}\prod_{i=1}^{j}\delta_{u_i}. \tag{69}$$

Proof. We prove the identity and the weight bound together by induction on t.

Base case and definition. Equation (68) gives $\iota(\delta_0) = Wy = W\alpha_0$ and $\Lambda(\delta_0) \le \tau$. Now suppose the two assertions are known below $t > 0$. Every exponent in (69) is nonnegative, and every coefficient on its right has already been constructed. The formula therefore defines $\delta_t \in \mathcal{O}$.

Image in the function field. For one tuple $\mathbf{u} \in \mathcal{P}^{\circ}_{s,j,t}$, write

$$Q_{s,j,\mathbf{u}} = \rho_{s,j}\eta^{\omega(\mathbf{u})}\prod_{i=1}^{j}\delta_{u_i}.$$

The induction hypothesis and (57) give

$$\iota(Q_{s,j,\mathbf{u}}) = c_{s,j}W^{d-j}\iota(\eta)^{\omega(\mathbf{u})}\prod_{i=1}^{j}\left(W\iota(\eta)^{e_{u_i}}\alpha_{u_i}\right).$$

Collecting the powers of W and $\iota(\eta)$, then using (65), gives

$$\iota(Q_{s,j,\mathbf{u}}) = c_{s,j}W^{d}\iota(\eta)^{e_t-1}\prod_{i=1}^{j}\alpha_{u_i}.$$

Summing over the retained tuples and using (63) yields

$$\iota(\delta_t) = -W^{d}\iota(\eta)^{e_t-1}N_t.$$

Using $\zeta\alpha_t = -N_t$ and $\iota(\eta) = W^{d-1}\zeta$ now gives

$$\iota(\delta_t) = W\iota(\eta)^{e_t}\alpha_t.$$

Weight. By Corollary 6.3 and Lemma 6.4 and the induction hypothesis,

$$\Lambda(Q_{s,j,\mathbf{u}}) \le \Lambda(\rho_{s,j}) + \omega(\mathbf{u})\mu + \sum_{i=1}^{j}(\tau + e_{u_i}\mu).$$

The definition of $\omega(\mathbf{u})$ makes the total exponent contribution from η equal to $e_t - 1$. Equation (61) contributes the last copy of μ. Hence every summand in (69) satisfies

$$\Lambda(Q_{s,j,\mathbf{u}}) \le \tau + e_t\mu.$$

The weight of a finite sum is at most the largest weight of its summands. This completes the induction. $\square$

The proof uses τ only in the monic reduction and in the coefficient estimate (61). No equality involving $\deg W + \ell$ enters the recurrence.

Proposition 6.6 (Relation with the original numerators)**.** *Let β_t be the corrected source numerator of Theorem 4.4. Then*

$$\delta_0 = \beta_0 = T, \qquad \delta_t = W^{t-1}\beta_t \quad (t > 0). \tag{70}$$

If $\beta_t \neq 0$, then

$$\Lambda(\delta_t) = (t-1)w + \Lambda(\beta_t). \tag{71}$$

Consequently (36) *and* (67) *have identical numerical content.*

Proof. The source numerator satisfies $\iota(\beta_t) = W^{t+1}\iota(\xi)^{e_t}\alpha_t$, while (66) and $\eta = W\xi$ give

$$\iota(\delta_t) = W^{e_t+1}\iota(\xi)^{e_t}\alpha_t.$$

For $t > 0$, the exponent difference is $e_t + 1 - (t+1) = t - 1$. Injectivity of ι proves (70). Multiplying a nonzero canonical representative by W^{t-1} leaves its T-degree unchanged. Thus no reduction modulo $\widehat{H}$ is required. Full additivity in $K[T, Z]$ therefore gives (71). Finally, $P_t + (t-1)w = \tau + e_t\mu$ by (37). □

7 From coefficient bounds to a curve of codewords

7.1 Counting exceptional challenge values

A resultant turns a two-variable root question into a one-variable degree bound. Suppose two polynomials in T share a root after we set $Z = z$. Then their resultant, viewed as a polynomial in Z, vanishes at that value of z. We shall therefore bound the degree of the resultant.

Let $u \in \mathcal{O}$ have canonical representative

$$P(T, Z) = \sum_{i=0}^{r} p_i(Z)T^i, \qquad r < h, \qquad \Lambda(P) \leq C. \tag{72}$$

Its controlling resultant is the polynomial in Z

$$S_u(Z) = \mathrm{Res}_T(P, \widehat{H}). \tag{73}$$

For a nonzero class u, the polynomial S_u is nonzero. Indeed, a zero resultant would mean that P and $\widehat{H}$ share a nonconstant factor over $K(Z)$. But $\widehat{H}$ is irreducible of degree h, while P has T-degree below h.

Lemma 7.1 (A degree bound for the resultant)**.** *The resultant in* (73) *satisfies*

$$\deg_Z S_u \leq hC.$$

Proof. The Sylvester matrix has h shifted rows of coefficients of P and r shifted rows of coefficients of $\widehat{H}$. Number the columns from 0 to $h + r - 1$. A nonzero determinant term chooses one coefficient from each row. Let $i_1, \ldots, i_h$ be the T-indices chosen in the P rows and $j_1, \ldots, j_r$ the indices chosen in the $\widehat{H}$ rows.

There is only one piece of bookkeeping. The chosen columns have total index $(h+r-1)(h+r)/2$. The shifts of the P-rows contribute $h(h-1)/2$, and the shifts of the $\widehat{H}$-rows contribute $r(r-1)/2$. Removing these shifts leaves

$$\sum_{a=1}^{h} i_a + \sum_{b=1}^{r} j_b = hr. \tag{74}$$

The coefficient bounds are

$$\deg p_i \le C - i\tau, \qquad \deg[T^j]\widehat{H} \le (h-j)\tau.$$

Thus the Z-degree of the determinant term is at most

$$hC + \tau\left(hr - \sum_a i_a - \sum_b j_b\right).$$

Equation (74) makes the term in parentheses zero. Every nonzero determinant term has degree at most hC, and the determinant itself has the same bound. $\square$

Corollary 7.2 (Zero count for a regular element)**.** *For $u \ne 0$, at most $h\Lambda(u)$ values $z \in K$ admit a value $t_z \in K$ with*

$$\widehat{H}(t_z, z) = 0, \qquad P(t_z, z) = 0.$$

Proof. Each such z is a root of S_u. Apply Lemma 7.1 and the univariate root bound. $\square$

7.2 Why the Hensel series truncates

We now use the zero count to show that the Hensel series is actually a polynomial of the required degree. Fix a surviving challenge z and let

$$t_z = W(z)p_z(x_0)$$

be the corresponding root of $\widehat{H}(T, z)$, and let $\pi_z : \mathcal{O} \to K$ be evaluation at (z, t_z). Write

$$a_{z,t} = [U^t]p_z(x_0 + U), \qquad \eta_z = \pi_z(\eta).$$

For every surviving challenge, $W(z) \ne 0$ and the chosen root is simple. Therefore $\eta_z \ne 0$.

Lemma 7.3 (Evaluation of the cleared coefficients)**.** *For every surviving selected challenge z and every $t \ge 0$,*

$$\pi_z(\delta_t) = W(z)\eta_z^{e_t} a_{z,t}. \tag{75}$$

Consequently, if every p_z has degree at most m, then $\pi_z(\delta_t) = 0$ for $t > m$.

Proof. At order zero, $\pi_z(T) = W(z)p_z(x_0) = W(z)a_{z,0}$. Suppose the identity is known below an order $t > 0$. For one tuple in $\mathcal{P}^{\circ}_{s,j,t}$, put

$$Q_{s,j,\mathbf{u}} = \rho_{s,j}\eta^{\omega(\mathbf{u})} \prod_{i=1}^{j} \delta_{u_i}.$$

The induction hypothesis gives

$$\pi_z(Q_{s,j,\mathbf{u}}) = c_{s,j}(z)W(z)^d \eta_z^{e_t - 1} \prod_{i=1}^{j} a_{z,u_i}.$$

The coefficient-t equation in $R(x_0+U, p_z(x_0+U), z) = 0$ is $\zeta_z a_{z,t} = -N_{z,t}$, where $N_{z,t}$ is the sum of the displayed nonlinear products and $\zeta_z = \partial_Y R(x_0, p_z(x_0), z)$. Since $\eta_z = W(z)^{d-1}\zeta_z$, applying π_z to (69) yields

$$\pi_z(\delta_t) = -W(z)^d \eta_z^{e_t-1} N_{z,t} = W(z)\eta_z^{e_t} a_{z,t}.$$

This proves (75). Only regular elements have been evaluated; no specialization map on the whole field L is needed. If $\deg p_z \le m$, then $a_{z,t} = 0$ for $t > m$, which gives the final assertion. ☐

Put

$$C_t = \tau + e_t\mu. \tag{76}$$

Let $D_X > m$ be the X-degree cutoff for the interpolation polynomial. Suppose the branch contains more than hC_t surviving challenges. For $t > m$, Lemma 7.3 makes δ_t vanish at every one of them. The zero count therefore gives $\delta_t = 0$ in $\mathcal{O}$. Since $W\iota(\eta)^{e_t}$ is nonzero, (66) gives $\alpha_t = 0$ in L.

Apply this argument for every $m < t < D_X$. The coefficients in that range vanish, so the finite sum

$$\gamma_m(X) = \sum_{t=0}^{m} \alpha_t (X - x_0)^t \tag{77}$$

is already an exact root. The Hensel equation says that the substituted parent vanishes modulo $(X - x_0)^{D_X}$. That substituted polynomial has X-degree below D_X, so divisibility by $(X - x_0)^{D_X}$ forces it to be zero. Uniqueness then identifies the formal lift with (77).

7.3 Clearing all remaining denominators at once

Truncation proves that the lift is a polynomial in X, but curve decodability also requires control of its dependence on Z. Claim 5.10 of the published proof uses Claim A.2 a second time for this purpose [BCIKS23, JACM p. 31:27]. It places all surviving coefficients over one common denominator. The cleared coefficients give the following numerator.

Lemma 7.4 (A common numerator for the truncated root)**.** *Assume $m \ge 1$ and set*

$$\Delta_m(X) := \sum_{t=0}^{m} \delta_t (X - x_0)^t \eta^{e_m - e_t} \in \mathcal{O}[X]. \tag{78}$$

Then

$$\iota(\Delta_m(X)) = W\iota(\eta)^{e_m} \gamma_m(X) \quad \textit{in } L[X]. \tag{79}$$

For every $x \in K$,

$$\Lambda(\Delta_m(x)) \le C_m. \tag{80}$$

Proof. Since e_t is nondecreasing, each exponent $e_m - e_t$ in (78) is nonnegative. Applying (66) to each summand gives

$$\iota(\delta_t)\iota(\eta)^{e_m - e_t} = W\iota(\eta)^{e_m}\alpha_t,$$

and summing proves (79). Evaluation at $x \in K$ has weight zero. By Corollary 6.3 and Theorem 6.5, each summand satisfies

$$\Lambda\big(\delta_t (x - x_0)^t \eta^{e_m - e_t}\big) \le \tau + e_t\mu + (e_m - e_t)\mu = C_m.$$

The addition inequality in Corollary 6.3 now gives (80). ☐

Let $v_x \in K^\times$ be the generalized Reed–Solomon multiplier at a fixed evaluation coordinate x; for an ordinary Reed–Solomon code take $v_x = 1$. Normalize the received words by

$$\widetilde{f}_j(x) = v_x^{-1} f_j(x)$$

and put

$$w_x(Z) = \sum_{j=0}^{M} \widetilde{f}_j(x) Z^j \in K[Z]. \tag{81}$$

Agreement of the encoded word $v_x p_z(x)$ with $f_z(x)$ is then equivalent to $p_z(x) = w_x(z)$.

In the parameterized-curve argument, Y is assigned the same weight as the degree M of w_x, so $\ell = M$ [BCIKS23, Sec. 6.2]. In the inseparable reduction the separable variable receives an even larger weight. In either case,

$$\deg w_x \le M \le \ell, \tag{82}$$

and this is exactly the weighting already used in the source arguments.

Define the integral discrepancy

$$E_x = \Delta_m(x) - W\eta^{e_m} w_x(Z) \in \mathcal{O}. \tag{83}$$

Lemma 7.5 (Recovering the value at one coordinate)**.** *Under* (82)*,*

$$\Lambda(E_x) \le C_m. \tag{84}$$

Let S_x be a set of surviving challenges for this branch such that $p_z(x) = w_x(z)$ for every $z \in S_x$. If

$$|S_x| > hC_m, \tag{85}$$

then

$$\gamma_m(x) = w_x(Z) \quad \textit{in } L. \tag{86}$$

Proof. The second term in (83) has weight at most

$$\deg W + e_m\mu + \deg w_x \le \deg W + e_m\mu + \ell \le \tau + e_m\mu = C_m,$$

using (9) and (82). In the saturated case $\deg W + \ell = \tau$, so the fit is exact. The same quantity τ that repairs the base coefficient also pays for the challenge polynomial. Together with (80), this proves (84).

For $z \in S_x$, Lemma 7.3 and the Taylor identity for the degree-m polynomial p_z give

$$\pi_z(\Delta_m(x)) = W(z)\eta_z^{e_m} \sum_{t=0}^{m} a_{z,t}(x - x_0)^t = W(z)\eta_z^{e_m} p_z(x).$$

Hence (83) specializes to zero because $p_z(x) = w_x(z)$. If $E_x \ne 0$, Corollary 7.2 and (84) permit at most hC_m such values of z, contradicting (85). Thus $E_x = 0$ in $\mathcal{O}$. Mapping to L and using (79) gives

$$W\iota(\eta)^{e_m}\big(\gamma_m(x) - w_x(Z)\big) = 0.$$

Both factors preceding the parenthesis are nonzero, so (86) follows. $\square$

In the notation of the source line proof, the common numerator is

$$\beta(X) = \sum_{t=0}^{m} \beta_t (X - x_0)^t W^{m-t} \xi^{e_m - e_t}. \tag{87}$$

Equation (78) is exactly $W^{m-1}\beta(X)$. Indeed, (70) and $\eta = W\xi$ give, for $t > 0$,

$$\delta_t \eta^{e_m - e_t} = W^{m-1} \beta_t W^{m-t} \xi^{e_m - e_t},$$

and the same identity holds at $t = 0$. The source discrepancy in the line case is

$$\beta(x) - w_x(Z) W^{m+1} \xi^{e_m}.$$

Multiplying it by W^{m-1} gives E_x, since $W\eta^{e_m} = W^{2m}\xi^{e_m}$. Thus the second resultant argument concerns the same root and the same values of z with $W(z) \neq 0$. Only the numerator used to represent it has changed.

7.4 Finding enough evaluation points

Let S be the surviving challenge set for the fixed branch, and let $A_z \subseteq D$ be the agreement support of p_z. Put

$$S_x = \{z \in S : x \in A_z\}.$$

The next count is independent of the algebra that produced the zero budget.

Lemma 7.6 (Finding $m + 1$ useful coordinates)**.** *Let E be a nonnegative integer. Suppose $|D| = n$, $|A_z| > \theta$ for every $z \in S$, and*

$$(\theta + 1)|S| > m|S| + nE. \tag{88}$$

Then at least $m + 1$ coordinates $x \in D$ satisfy $|S_x| > E$.

Proof. Count the pairs (x, z) for which $x \in A_z$. There are at least $(\theta + 1)|S|$. If at most m coordinates satisfied $|S_x| > E$, their total contribution would be at most $m|S|$, and every remaining coordinate would contribute at most E. The total would then be at most $m|S| + nE$, contrary to (88). □

Apply the lemma with $E = hC_m$, and choose distinct useful coordinates $x'_0, \ldots, x'_m$. For each $j = 0, \ldots, M$, let $P_j(X) \in K[X]$ be the polynomial of degree at most m interpolating the normalized values $\widetilde{f}_j(x'_i)$. By Lemma 7.5,

$$\gamma_m(x'_i) = \sum_{j=0}^{M} Z^j \widetilde{f}_j(x'_i) = \sum_{j=0}^{M} Z^j P_j(x'_i) \qquad (0 \leq i \leq m).$$

Both sides are polynomials in X over L of degree at most m. Therefore

$$\gamma_m(X) = \sum_{j=0}^{M} Z^j P_j(X). \tag{89}$$

It remains to show that the same identity holds for every surviving challenge. We can do this without defining a specialization of the whole field L. For every surviving z, the equality $E_{x'_i} = 0$ and Lemma 7.3 imply

$$p_z(x'_i) = w_{x'_i}(z) = \sum_{j=0}^{M} z^j \widetilde{f}_j(x'_i) = \sum_{j=0}^{M} z^j P_j(x'_i) \qquad (0 \leq i \leq m),$$

after cancelling the nonzero scalar $W(z)\eta_z^{e_m}$. The two sides are polynomials in X of degree at most m and agree at $m+1$ distinct points. Thus

$$p_z(X) = \sum_{j=0}^{M} z^j P_j(X).$$

For a generalized Reed–Solomon code, multiplying by v_x gives the encoded component words $v_x P_j(x)$ and hence the required degree-M curve of codewords.

The next theorem collects the argument for one fixed branch. Its list of hypotheses is long because it states every denominator, degree, and simple-root condition explicitly. The conclusion is much shorter: all codewords assigned to the branch lie on one degree-M curve.

Theorem 7.7 (From one branch to one curve of codewords)**.** *Let $S \subseteq K$ be finite. For each $z \in S$, let $p_z \in K[X]$ have degree at most $m \geq 1$ and assume*

$$H(p_z(x_0), z) = 0, \qquad R(X, p_z(X), z) = 0, \qquad W(z) \neq 0. \tag{90}$$

Put $t_z = W(z)p_z(x_0)$. The first equality in (90) *gives $\widehat{H}(t_z, z) = 0$, so evaluation at (z, t_z) defines*

$$\pi_z : \mathcal{O} \longrightarrow K, \qquad Z \longmapsto z, \qquad T \longmapsto t_z. \tag{91}$$

Assume also that

$$\pi_z(\eta) \neq 0 \qquad (z \in S). \tag{92}$$

Let $A_z \subseteq D$ satisfy

$$p_z(x) = w_x(z) \qquad (x \in A_z),$$

where $w_x \in K[Z]$ has degree at most $M \leq \ell$. Assume that

$$|S| > hC_t \qquad \text{for every } t \text{ with } m < t < D_X, \tag{93}$$

that the interpolation degree bound gives $\deg_X R(X, \gamma_m(X), Z) < D_X$, and that there is a set $D_{\mathrm{top}} \subseteq D$ with

$$|D_{\mathrm{top}}| = m+1, \qquad |S_x| > hC_m \quad (x \in D_{\mathrm{top}}). \tag{94}$$

Then there are polynomials $P_0, \ldots, P_M \in K[X]$ of degree at most m such that

$$p_z(X) = \sum_{j=0}^{M} z^j P_j(X) \qquad (z \in S). \tag{95}$$

For a generalized Reed–Solomon code, the corresponding component codewords are $(v_x P_j(x))_{x \in D}$.

Proof. For $m < t < D_X$, Lemma 7.3 makes every specialization of δ_t on S vanish. The cardinality assumption (93) and Corollary 7.2 therefore give $\delta_t = 0$ in $\mathcal{O}$. Equation (66), injectivity of ι, and the nonvanishing of $W\iota(\eta)^{e_t}$ give $\alpha_t = 0$. The degree bound then shows that the truncated series γ_m is an actual polynomial root, by the argument in Section 7.

For every $x \in D_{\mathrm{top}}$, Lemma 7.5 gives $\gamma_m(x) = w_x(Z)$. Choose $m+1$ distinct coordinates from D_{top} and interpolate the normalized coefficient values $\widetilde{f}_j(x)$ to obtain P_j of degree at most m. The two sides of (89) have X-degree at most m and agree at those coordinates, so they are equal. Finally, specialize the integral identities at an arbitrary $z \in S$ and cancel the nonzero factor $W(z)\eta_z^{e_m}$. This proves (95). Multiplying by the coordinate multipliers returns the corresponding generalized Reed–Solomon codewords. □

The theorem is deliberately stated for $m \geq 1$, which is the range used in the proximity-gap arguments. The constant-polynomial case can be treated separately and is not needed here.

Corollary 7.8 (A convenient counting condition)**.** *Under the remaining hypotheses of Theorem 7.7, suppose $|D| = n$, $|A_z| > \theta$ for every $z \in S$, and* (88) *holds with $E = hC_m$. Then the conclusion of Theorem 7.7 holds.*

Proof. Apply Lemma 7.6 with $E = hC_m$. At least $m+1$ coordinates satisfy $|S_x| > hC_m$. Choose any $m+1$ of them to form a set D_{top} satisfying (94). □

7.5 Why the original numerical bounds still suffice

We have repaired the local estimate. We must still check that the repaired number fits inside the allowance used by the published proof.

Let D_* be a common upper bound for D_H and D_R. Assume $1 \leq h \leq d$, $D_H \geq \ell h$, and $D_R \geq \ell$. For every $\ell \geq 1$,

$$\tau = D_H + \ell - \ell h \leq D_*, \tag{96}$$

$$\mu = (D_R - \ell) + (d-1)(D_H - \ell h) \leq dD_*. \tag{97}$$

The first inequality uses $h \geq 1$; the second follows by discarding the two nonpositive subtraction terms.

Claim A.2 displays three successive quantities. Only the last one is used by the later root counts. The first is the parametric expression

$$B_t^{\text{par}} = 1 + (t+1)\deg W + e_t\Lambda(\xi). \tag{98}$$

The paper next bounds this by the numerical expression

$$B_t^{\text{par}} \leq B_t^{\text{num}} := 1 + (t+1)(D_* - h) + e_t(d-1)(D_* - h + 1), \tag{99}$$

and finally proves

$$B_t^{\text{num}} < (2t+1)dD_*. \tag{100}$$

The unsupported step is the derivation of (98). The later truncation and common-numerator arguments use only the coarser quantity in (100) [BCIKS23, Secs. 5.2.6–5.2.7].

The direct and denominator-free bounds are

$$P_t = \tau + tw + e_t\sigma, \qquad C_t = \tau + e_t\mu = P_t + (t-1)w \quad (t > 0).$$

The repaired bounds need not reproduce the middle line B_t^{num}. Now specialize to the line case, where $D_R = D_H = D_*$ and $\ell = 1$. The differences from the middle line are

$$P_t - B_t^{\text{num}} = (t-1)(D_* - h - w) - (2t-1)(d-h), \tag{101}$$

$$C_t - B_t^{\text{num}} = (t-1)(D_* - h) - (2t-1)(d-h). \tag{102}$$

The first expression is nonpositive in the saturated case $w = D_* - h$. Either difference may become positive when the leading coefficient has smaller degree. That is precisely the case missed by the printed estimate. It does not affect the proof, because the later zero counts use the final coarse allowance. From (96)–(97), for every $t > 0$,

$$hC_t \leq h\bigl(D_* + (2t-1)dD_*\bigr) \leq 2thdD_* < h(2t+1)dD_*. \tag{103}$$

Thus the repaired estimate may exceed the middle line of the printed chain, but it remains strictly below the threshold actually used by both zero-count arguments.

Corollary 7.9 (Application to the published line argument)**.** *Consider the line case $M = \ell = 1$ under the hypotheses of Claims 5.8–5.11 of [BCIKS23]. After the identifications $k = m$, $d_H = h$, and $D = D_*$, the repaired coefficients establish the truncation conclusion of Claim 5.8 and the local exactness conclusion of Claim 5.10 without increasing either exceptional-set allowance.*

Proof. For every $m < t < D_X$, Claim 5.8 compares the surviving branch set with $h(2t + 1)dD_*$. Equation (103) gives $hC_t < h(2t + 1)dD_*$, so (93) holds.

Claim 5.11 supplies a set D_{top} of $m + 1$ coordinates. Its proof first establishes

$$|S_x| \geq \frac{1-\rho-\delta}{1-\rho}|S'| \qquad (x \in D_{\mathrm{top}})$$

and then uses the source parameter inequality to conclude

$$|S_x| > (2m + 1)hdD_* \qquad (x \in D_{\mathrm{top}}). \tag{104}$$

At $t = m$, (103) gives $hC_m < h(2m + 1)dD_*$. Thus (104) implies (94). Apply Theorem 7.7. $\square$

The parameterized-curve extension of the source assigns Y weight $\ell = M$, so (82) holds by construction [BCIKS23, Sec. 6.2]. Appendix C treats a factor $R(X, Y^q, Z)$, where $q = p^f$, by using the separable variable $\widetilde{Y} = Y^q$ with weight $\ell = q$. The local estimates above therefore apply there as well. The remaining question is global: can the required degree bounds still be summed over all factors? We answer that next.

7.6 The degree bookkeeping in the 2026 refinement

The 2026 refinement reuses the Hensel-and-resultant argument above [BCHKS26, Sec. 3.2, pp. 24–25]. Its improved bound depends on a careful sum over the irreducible factors of the interpolation polynomial. There is one extra point to check: the degree used for a factor must control all coefficients that appear after shifting X.

7.6.1 Why the specialized degree is not enough

Lemma 3.1 of the refinement bounds the full (Y, Z)-weighted degree of the interpolant [BCHKS26, pp. 21–22]. Later, the abbreviated factor count uses a degree $D_Z^{(R_i)}$ taken from the content-free specialization $R_i(x_0, Y, Z)$. That degree controls the coefficient of $(X - x_0)^0$, but it need not control the higher shifted coefficients. For example,

$$R_N(X, Y, Z) = Y + Z + XZ^N, \qquad x_0 = 0, \tag{105}$$

has content-free specialization $Y + Z$ of weighted degree 1, while the coefficient of $X - x_0$ is Z^N. Thus one cannot set $D_R = D_Z^{(R_i)}$ without a further argument.

The remedy is to use the full weighted degree

$$G_i = \operatorname{wdeg}_{Y,Z} R_i(X, Y, Z), \tag{106}$$

where X has weight zero and Y, Z have weight one. By Lemma 4.1, every shifted coefficient of R_i satisfies

$$\deg c_{s,j} + j \leq G_i. \tag{107}$$

For (105), the correct value is $G_i = N$.

7.6.2 Separable factors

We first treat the separable factorization used in the main body of the refinement.

Proposition 7.10 (Full-degree factor summation)**.** *Let $D_X, D_Y, D_Z, B \in \mathbb{N}_{>0}$. Suppose $Q \neq 0$ and that the separable part of the factorization in Section 3.2 of [BCHKS26] is*

$$Q(X,Y,Z) = C(X,Z)\prod_i R_i(X,Y,Z)^{\nu_i}, \qquad C \neq 0, \qquad \nu_i \geq 1. \tag{108}$$

Assume that every R_i is irreducible, separable in Y, and nonconstant in Y. Assume that Q has Y-degree at most D_Y and (Y,Z)-weighted degree at most D_Z, with X assigned weight zero. Put

$$d_i = \deg_Y R_i, \qquad G_i = \operatorname{wdeg}_{Y,Z} R_i, \qquad c_0 = \deg_Z C. \tag{109}$$

Choose x_0 as in the refinement so that every positive-Y-degree specialization $R_i(x_0,Y,Z)$ is separable over $K(Z)$, and write

$$R_i(x_0,Y,Z) = C_i(Z)\prod_{j=1}^{r_i} H_{ij}(Y,Z), \qquad C_i \in K[Z]\setminus\{0\}, \qquad h_{ij} = \deg_Y H_{ij}, \tag{110}$$

where the polynomials H_{ij} are the nonconstant irreducible factors, and W_{ij} is the leading coefficient of H_{ij} in Y. Let S be a finite set of challenges such that for every $z \in S$ there is a polynomial p_z with $Q(X,p_z(X),z) = 0$. If

$$|S| > 2D_X D_Y^2 D_Z + BD_Y, \tag{111}$$

then there are indices i,j and a set $S'_{ij} \subseteq S$ such that every shifted coefficient of R_i is bounded by G_i and

$$|S'_{ij}| > (2D_X - 1)d_i h_{ij} G_i + B. \tag{112}$$

Moreover, for every $z \in S'_{ij}$,

$$\begin{aligned} R_i(X,p_z(X),z) &= 0, & H_{ij}(p_z(x_0),z) &= 0,\\ C_i(z) &\neq 0, & W_{ij}(z) &\neq 0,\\ \partial_Y R_i(x_0,p_z(x_0),z) &\neq 0. \end{aligned} \tag{113}$$

Thus the same global threshold supplies the local degree and cardinality bounds required by the corrected Hensel argument.

Proof. We divide the count into four steps.

1. Global degree bounds. Weighted degree is additive under multiplication because the coefficient ring is an integral domain. Ordinary degree in Y is additive for the same reason. Hence

$$c_0 + \sum_i \nu_i G_i \leq D_Z, \qquad \sum_i \nu_i d_i \leq D_Y. \tag{114}$$

Translation in the weight-zero variable X preserves the full bound. In particular, if

$$R_i(x_0+U,Y,Z) = \sum_{s,j} c_{s,j}^{(i)}(Z)U^s Y^j,$$

then

$$\deg c_{s,j}^{(i)} + j \leq G_i \tag{115}$$

whenever the coefficient is nonzero. This is Lemma 4.1 with $\ell = 1$.

2. Content factors and local branches. Specializing $X = x_0$ cannot increase either weighted degree or Y-degree. Therefore

$$\sum_{j=1}^{r_i} h_{ij} \le d_i, \qquad r_i \le d_i, \qquad \deg C_i \le G_i. \tag{116}$$

Each factor H_{ij} also has weighted degree at most G_i.

The factors C and C_i collect terms that are independent of the response variable Y. A zero can therefore come from a content factor rather than from one of the displayed R_i or H_{ij}. The refinement points this out explicitly and identifies it as a minor omission in the earlier proof [BCHKS26, p. 25, n. 5]. We now include those values in the count.

Let I_+ be the set of indices with $r_i > 0$. If $r_i = 0$, then $R_i(x_0, Y, Z) = C_i(Z)$ is independent of Y. A successful challenge assigned to this factor must therefore be a root of C_i. Discard those challenges, together with the values for which $C(X, z)$ is the zero polynomial in X. The latter set has at most c_0 elements, since it is contained in the roots of any nonzero coefficient of C, viewed as a polynomial in X. Their degrees must be charged to the global budget. The total number discarded is therefore at most

$$c_0 + \sum_{i \notin I_+} \deg C_i \le c_0 + \sum_{i \notin I_+} \nu_i G_i. \tag{117}$$

The second inequality follows because C_i divides the specialization of R_i, and specialization cannot increase weighted degree.

3. Assigning challenges to factor pairs. For each remaining z, evaluate (108) at $Y = p_z(X)$. The product vanishes in the integral domain $K[X]$, so at least one factor vanishes. Choose an index i such that $R_i(X, p_z(X), z) = 0$. Necessarily $i \in I_+$. If $C_i(z) \ne 0$, choose a factor H_{ij} that vanishes at $p_z(x_0)$. If $C_i(z) = 0$, assign z temporarily to any one of the r_i factors. Make each choice once, and call the resulting disjoint sets $\widetilde{S}_{ij}$.

We now show that one pair receives many challenges. Suppose, to the contrary, that every pair with $i \in I_+$ satisfies

$$|\widetilde{S}_{ij}| \le 2D_X d_i h_{ij} G_i + B. \tag{118}$$

Using (117) and summing over the pairs gives

$$|S| \le c_0 + \sum_{i \notin I_+} \nu_i G_i + 2D_X \sum_{i \in I_+} d_i^2 G_i + B \sum_{i \in I_+} r_i. \tag{119}$$

Since $d_i \le D_Y$, $\nu_i \ge 1$, and (116) holds, we have

$$\sum_{i \in I_+} d_i^2 G_i \le D_Y^2 \sum_{i \in I_+} \nu_i G_i, \qquad \sum_{i \in I_+} r_i \le D_Y. \tag{120}$$

Put $A = 2D_X D_Y^2$. Since $A \ge 1$, equations (114)–(120) give

$$\begin{aligned} |S| &\le c_0 + \sum_{i \notin I_+} \nu_i G_i + A \sum_{i \in I_+} \nu_i G_i + BD_Y, \\ |S| &\le A \left(c_0 + \sum_i \nu_i G_i \right) + BD_Y, \\ |S| &\le 2D_X D_Y^2 D_Z + BD_Y. \end{aligned} \tag{121}$$

This contradicts (111). Some pair therefore satisfies

$$|\widetilde{S}_{ij}| > 2D_X d_i h_{ij} G_i + B. \tag{122}$$

4. Removing temporary content assignments. Some challenges were assigned to a branch only because $C_i(z) = 0$. We must remove them without spending more than the local pole allowance. Fix the large pair found above, write $W = W_{ij}$, and expand

$$R_i(x_0, Y, Z) = \sum_{r=0}^{d_i} b_r(Z) Y^r.$$

The distinction between a polynomial and its quotient class matters here. We therefore begin with the raw polynomial numerator

$$N_{ij}(T, Z) = \sum_{r=1}^{d_i} r\, b_r(Z) W^{d_i - r} T^{r-1} \in K[Z, T]. \tag{123}$$

In $K(Z)[T]$, this polynomial equals $W^{d_i-1} \partial_Y R_i(x_0, T/W, Z)$. Every term of N_{ij} is divisible by W. This is immediate when $r < d_i$. When $r = d_i$, it follows because the leading coefficient $b_{d_i} = C_i \prod_j \mathrm{lc}_Y(H_{ij})$ contains the leading coefficient W of the chosen factor. We may therefore define

$$\Xi_{ij} = N_{ij}/W \in K[Z, T], \qquad N_{ij} = W \Xi_{ij}. \tag{124}$$

Only now do we pass to the quotient. Let $\xi_{ij}(T, Z)$ be the remainder of Ξ_{ij} upon division by the monic polynomial $\widehat{H}_{ij}$. Its residue class is the regular derivative numerator used in the local argument.

The factorization in (110) gives $C_i \mid \partial_Y R_i(x_0, Y, Z)$, and hence $N_{ij} = C_i A_{ij}$ for some $A_{ij} \in K[Z, T]$. Remainder modulo a monic polynomial is $K[Z]$-linear, so

$$W \xi_{ij} = N_{ij} \bmod \widehat{H}_{ij} = C_i (A_{ij} \bmod \widehat{H}_{ij}). \tag{125}$$

In particular,

$$C_i(Z) \mid W(Z) \xi_{ij}(T, Z) \qquad \text{in } K[Z, T]. \tag{126}$$

Since $R_i(x_0, Y, Z)$ is separable in Y, its derivative is nonzero at the generic root of H_{ij}. Thus the class of ξ_{ij} is nonzero in the branch field, and $\mathrm{Res}_T(\xi_{ij}, \widehat{H}_{ij})$ is a nonzero polynomial.

If $C_i(z) = 0$, equation (126) says that $W(z)\xi_{ij}(T, z)$ is the zero polynomial in T. Thus either $W(z) = 0$ or $\xi_{ij}(T, z)$ is identically zero. Every temporarily assigned challenge is therefore a root of

$$W(Z)\, \mathrm{Res}_T(\xi_{ij}, \widehat{H}_{ij}). \tag{127}$$

The pole estimate (46), with $D_* = G_i$, shows that this polynomial has degree strictly less than $d_i h_{ij} G_i$. Removing all its roots from (122) leaves more than

$$(2D_X - 1) d_i h_{ij} G_i + B \tag{128}$$

surviving specializations.

The selected pair now has exactly the data needed by the local Hensel argument. Equation (115) controls every shifted coefficient, and H_{ij} has weighted degree at most G_i. For each positive coefficient index $t < D_X$, and for the common-numerator index $m < D_X$, the repaired zero-count bound is strictly below $(2D_X - 1) d_i h_{ij} G_i$; compare (103). The reserve B is left for the final collinearity step.

Let S'_{ij} be the surviving set. By construction, $R_i(X, p_z(X), z) = 0$ and $H_{ij}(p_z(x_0), z) = 0$ for every $z \in S'_{ij}$. All roots of the polynomial in (127) were removed, so $W_{ij}(z) \neq 0$ and $\mathrm{Res}_T(\xi_{ij}, \widehat{H}_{ij})(z) \neq 0$. The latter says that $\xi_{ij}(T, z)$ does not vanish at the branch root of $\widehat{H}_{ij}(T, z)$; since $W_{ij}(z) \neq 0$, this is equivalent to $\partial_Y R_i(x_0, p_z(x_0), z) \neq 0$. Equation (126) also shows that $C_i(z) \neq 0$. Thus (113) holds, and the proposition follows. $\square$

7.6.3 Inseparable factors

The remaining case occurs when a global factor depends on a Frobenius power of Y. There are two changes to keep track of. First, the powered candidate has a larger degree in X. Second, after taking a Frobenius root, the second resultant must be applied to a common numerator built from all retained coefficients, just as in the separable proof. We spell out both points.

Corollary 7.11 (Inseparable factors)**.** *Retain the positive integers D_X, D_Y, D_Z, B and the challenge data of Proposition 7.10. Let K be a finite field of characteristic $p > 0$, and suppose*

$$Q(X,Y,Z) = C(X,Z)\prod_i R_i(X, Y^{q_i}, Z)^{\nu_i}, \qquad q_i = p^{f_i}, \tag{129}$$

where $Q \neq 0$, $C \neq 0$, $f_i \geq 0$, $\nu_i \geq 1$, and each $R_i(X, \widetilde{Y}, Z)$ is irreducible, separable, and nonconstant in $\widetilde{Y}$. Give $\widetilde{Y}$ weight q_i and Z weight one, and put

$$d_i = \deg_{\widetilde{Y}} R_i, \qquad G_i = \operatorname{wdeg}_{(q_i,1)} R_i. \tag{130}$$

Choose x_0 so that every positive-$\widetilde{Y}$-degree specialization $R_i(x_0, \widetilde{Y}, Z)$ is separable over $K(Z)$. Then

$$c_0 + \sum_i \nu_i G_i \leq D_Z, \qquad \sum_i \nu_i q_i d_i \leq D_Y, \tag{131}$$

and the threshold (111) *selects indices i, j and a surviving set S'_{ij} for which*

$$R_i(X, p_z(X)^{q_i}, z) = 0, \qquad H_{ij}(p_z(x_0)^{q_i}, z) = 0 \quad (z \in S'_{ij}), \tag{132}$$

together with

$$C_i(z) \neq 0, \qquad W_{ij}(z) \neq 0, \qquad \partial_{\widetilde{Y}} R_i(x_0, p_z(x_0)^{q_i}, z) \neq 0. \tag{133}$$

Fix such a pair and write $q = q_i$, $h = h_{ij}$. Thus the local branch weight is $\ell = q = q_i = p^{f_i}$. Let $k \geq 1$, and let $\mathcal{D}$ be an evaluation set of size n. Suppose that $\deg p_z \leq k$ *for $z \in S'_{ij}$, and that there are functions $u, v : \mathcal{D} \to K$ and supports $A_z \subseteq \mathcal{D}$ such that*

$$|A_z| > \theta, \qquad p_z(x) = u_x + v_x z \quad (x \in A_z). \tag{134}$$

Assume also that Q has $(1, k, 0)$-weighted degree below D_X. For this local argument take

$$D_R = D_H = G_i, \qquad b_H = G_i - qh, \qquad \tau = b_H + q, \qquad \mu = (G_i - q) + (d_i - 1)b_H. \tag{135}$$

Retain the notation

$$C_t = \tau + e_t \mu \tag{136}$$

from (76)*. If*

$$(\theta + 1)|S'_{ij}| > k|S'_{ij}| + nhC_{qk}, \tag{137}$$

then there are polynomials $P_0, P_1 \in K[X]$, each of degree at most k, such that

$$p_z(X) = P_0(X) + zP_1(X) \qquad (z \in S'_{ij}). \tag{138}$$

The same global exceptional-set allowance as in the separable case is therefore sufficient.

Proof. We proceed in five steps. The first is the global count. The remaining four explain how the powered branch is converted back into the required line.

1. Global degree bookkeeping. After substituting $\widetilde{Y} = Y^{q_i}$, the monomials $\widetilde{Y}^r Z^s$ and $Y^{q_i r} Z^s$ both have weight $q_i r + s$. Additivity of weighted degree and ordinary Y-degree in (129) gives (131). Translation in the weight-zero variable X also gives the coefficient bound needed by the local Hensel argument. If

$$R_i(x_0 + U, \widetilde{Y}, Z) = \sum_{s,j} c^{(i)}_{s,j}(Z) U^s \widetilde{Y}^j,$$

then every nonzero coefficient satisfies

$$\deg c^{(i)}_{s,j} + q_i j \le G_i. \tag{139}$$

In particular,

$$\sum_i r_i \le \sum_i d_i \le \sum_i \nu_i q_i d_i \le D_Y, \tag{140}$$

and, since $d_i \le q_i d_i \le D_Y$ and $\nu_i \ge 1$,

$$\sum_i d_i^2 G_i \le D_Y^2 \sum_i \nu_i G_i. \tag{141}$$

These are the two estimates used in the factor-pair count of Proposition 7.10. The content polynomials still lie in $K[Z]$, and the raw derivative numerator is now formed by differentiating with respect to $\widetilde{Y}$. The same resultant therefore removes their roots. We obtain the stated pair, surviving set, and simple branch data.

2. Reindexing the challenges and bounding substitution degree. A finite field is perfect. We use this twice: first to invert Frobenius on the challenge set, and later to take coefficientwise qth roots. The map

$$\varphi_q : K \longrightarrow K, \qquad \varphi_q(r) = r^q,$$

is therefore a bijection. Define

$$\widetilde{S} = \varphi_q^{-1}(S'_{ij}), \qquad \widetilde{p}_r = p_{r^q}, \qquad \widetilde{A}_r = A_{r^q}. \tag{142}$$

Then $|\widetilde{S}| = |S'_{ij}|$, and the support sizes and all support-incidence counts are unchanged. Equations (132) become

$$R_i(X, \widetilde{p}_r(X)^q, r^q) = 0, \qquad H_{ij}(\widetilde{p}_r(x_0)^q, r^q) = 0. \tag{143}$$

The factor $R_i(X, Y^q, Z)$ has $(1, k, 0)$-weighted degree below D_X. Equivalently, $R_i(X, \widetilde{Y}, Z)$ has $(1, qk, 0)$-weighted degree below D_X. Thus $qk < D_X$.

Now let $\Gamma \in K[X]$ have degree at most qk. A monomial $X^b \widetilde{Y}^r Z^s$ contributes X-degree at most $b + r \deg \Gamma$, which is at most $b + rqk$. The weighted-degree hypothesis puts every such number below D_X. Taking the maximum over the monomials of R_i gives

$$\deg_X R_i(X, \Gamma(X), Z) < D_X. \tag{144}$$

This is the degree estimate needed to turn a truncated Hensel solution into an exact polynomial root.

The powered candidate $\widetilde{p}_r^q$ has degree at most qk. In characteristic p, its Taylor series around x_0 has nonzero coefficients only at indices divisible by q. Every coefficient to be killed has a positive index $t < D_X$. Applying (45) with $D_* = G_i$ gives

$$hC_t < h(2t+1) d_i G_i \le (2D_X - 1) d_i h G_i. \tag{145}$$

The first resultant therefore kills every coefficient whose index is not divisible by q, and every coefficient above qk, as long as the index is below D_X. By (144), the resulting sparse polynomial is the exact lifted root.

3. Taking a compatible Frobenius root. Let $\mathcal{O}$ be the regular quotient of the powered branch. Introduce rooted variables $\widetilde{T}, \widetilde{Z}$ with

$$\widetilde{T}^q = T, \qquad \widetilde{Z}^q = Z, \tag{146}$$

and let $\widetilde{\mathcal{O}}$ be the regular quotient of the coefficientwise inverse-Frobenius branch. Coefficientwise inverse Frobenius sends T to $\widetilde{T}$ and Z to $\widetilde{Z}$, while taking qth roots of coefficients. It gives a ring isomorphism

$$\widetilde{\sigma} : \mathcal{O} \xrightarrow{\sim} \widetilde{\mathcal{O}}.$$

Since both rings are integral domains, this isomorphism extends uniquely to their fraction fields. It commutes with monic reduction. Hence canonical representatives have the same exponent support before and after applying $\widetilde{\sigma}$. If $\widetilde{\Lambda}$ is the rooted weight with

$$\widetilde{\Lambda}(\widetilde{T}) = \Lambda(T), \qquad \widetilde{\Lambda}(\widetilde{Z}) = 1,$$

then

$$\widetilde{\Lambda}(\widetilde{\sigma}(a)) = \Lambda(a) \qquad (a \in \mathcal{O}). \tag{147}$$

The same transport preserves the nonvanishing needed later. If a rooted leading-coefficient value or rooted derivative value were zero, then its qth power would be the corresponding source value. The source value is nonzero by (133), so this is impossible in a field.

The sparse powered lift consequently has a unique qth root $\bar{\gamma}(X)$ of degree at most k. Its specialization at a rooted challenge $r \in \widetilde{S}$ is $\widetilde{p}_r = p_{r^q}$.

4. The rooted common numerator and the second resultant. A bound for only the highest Hensel coefficient would not be enough here. As in Lemma 7.4, the second resultant requires one numerator containing every retained coefficient. Define

$$\widetilde{\eta} = \widetilde{\sigma}(\eta), \qquad \widetilde{\delta}_t = \widetilde{\sigma}(\delta_{qt}) \quad (0 \le t \le k). \tag{148}$$

Put $g_t = e_{qk} - e_{qt}$. Write $\widetilde{\Delta}_k(X)$ for the polynomial

$$\sum_{t=0}^{k} \widetilde{\delta}_t \, \widetilde{\eta}^{g_t} (X - x_0)^t. \tag{149}$$

The exponents g_t are nonnegative. By (147) and Theorem 6.5,

$$\widetilde{\Lambda}(\widetilde{\delta}_t) \le \tau + e_{qt}\mu, \qquad \widetilde{\Lambda}(\widetilde{\eta}) \le \mu.$$

Therefore every summand in (149) has weight at most C_{qk}, and

$$\widetilde{\Lambda}(\widetilde{\Delta}_k(x)) \le C_{qk} \qquad (x \in K). \tag{150}$$

The image of $\widetilde{\Delta}_k$ in the rooted branch field is

$$\widetilde{W} \, \widetilde{\eta}^{e_{qk}} \bar{\gamma}(X),$$

where $\widetilde{W} = \widetilde{\sigma}(W)$. Thus $\widetilde{\Delta}_k$ clears the whole rooted polynomial $\bar{\gamma}$, not merely its top coefficient.

At a coordinate $x \in \mathcal{D}$, the received polynomial in the rooted challenge variable is

$$\widetilde{w}_x(\widetilde{Z}) = u_x + v_x \widetilde{Z}^q. \tag{151}$$

It has degree q, which is exactly the branch weight ℓ. Define the rooted discrepancy

$$\widetilde{E}_x = \widetilde{\Delta}_k(x) - \widetilde{W}\, \widetilde{\eta}^{e_{qk}} \widetilde{w}_x(\widetilde{Z}). \tag{152}$$

The same calculation as in Lemma 7.5 gives

$$\widetilde{\Lambda}(\widetilde{E}_x) \leq C_{qk}. \tag{153}$$

For $r \in \widetilde{S}$ and $x \in \widetilde{A}_r$, equation (134) gives

$$\widetilde{p}_r(x) = p_{r^q}(x) = u_x + v_x r^q = \widetilde{w}_x(r),$$

so $\widetilde{E}_x$ specializes to zero.

Because Frobenius preserves the incidence data, apply Lemma 7.6 with $m = k$ and $E = hC_{qk}$. Equation (137) then gives at least $k+1$ coordinates at which more than hC_{qk} rooted challenges agree. At each such coordinate the weighted resultant and (153) force $\widetilde{E}_x = 0$. The required local budget fits inside the surviving factor-pair allowance because

$$hC_{qk} < h(2qk+1)d_i G_i \leq (2D_X - 1)d_i h G_i. \tag{154}$$

5. Interpolation and return to the original challenge. Interpolate the two coefficient functions at the $k+1$ heavy coordinates. This gives $P_0, P_1 \in K[X]$ of degree at most k and

$$\bar{\gamma}(X) = P_0(X) + \widetilde{Z}^q P_1(X) \quad \text{in the rooted branch field.} \tag{155}$$

Now take $z \in S'_{ij}$ and let

$$r = \varphi_q^{-1}(z).$$

Then $r^q = z$, and the specialization of (155) at $\widetilde{Z} = r$ gives

$$p_z = \widetilde{p}_r = P_0 + r^q P_1 = P_0 + z P_1.$$

This is (138). Frobenius is a bijection, so no challenge and no support incidence is lost in the reindexing. □

We may now summarize the 2026 correction. The degree of the content-free specialization does not control the shifted coefficients, but the full degrees G_i do. Lemma 3.1 of the refinement supplies the global weighted bound on Q. The proposition and corollary above show that the full factor degrees can be summed without changing the threshold $2D_X D_Y^2 D_Z + BD_Y$. In the notation of the refinement, $B = \gamma n + 1$.

Curve-decodability statements of this form also occur as soundness interfaces in recent proof-system specifications [Cetal26].

8 Supplementary formal verification

The proof above does not depend on a proof assistant. The correction, the two resultant arguments, the factor-degree transfer, and the inseparable Frobenius argument are all proved in the manuscript. A separate Lean 4 development provides an independent check.

The canonical archival artifact for Paper Version 1 is the annotated release `paper-v1.0.1` in this repository:

`https://github.com/DJBarker87/weighted-hensel-repair`.

The release tag pins the exact artifact commit, and Zenodo archives the release series under the stable concept DOI `https://doi.org/10.5281/zenodo.22180553`. The mathematical Lean sources checked here are unchanged from commit

`f237d80eafdfbb8fdf5a3464e65510c824452bd5`.

That commit is green. The command `./scripts/verify.sh` performs the canonical clean replay.

The development checks the counterexamples, both corrected Hensel recurrences, their equivalence, the weighted resultant bound, both uses of the resultant, fixed-branch completion, the separable full-factor argument, and the inseparable Frobenius extension. Its main entry point also contains two application-specific numerical instances; those are supplementary and are not used by any theorem in this paper.

The generic formal theorems expose the same mathematical data as hypotheses: factorizations, branch choices, degree bounds, specialized separability, nonvanishing conditions, agreement sets, and cardinality inequalities. They do not claim an algorithm for discovering a factorization or an executable Guruswami–Sudan decoder. The sources depend only on Lean 4 and Mathlib, contain no admitted propositions or project-specific axioms, and report only *propext*, *Classical.choice*, and *Quot.sound*. A statement-by-statement manuscript alignment accompanies the tagged source package.

9 Conclusion

Claim A.2 underestimates the weight of the Hensel numerators when the leading coefficient of the chosen branch does not attain its permitted degree. The failure is visible at order zero, in the auxiliary numerators, and in the stated bound for the derivative numerator ξ. The later expression $\Lambda(\alpha_t) = \Lambda(Y) = 1$ also applies the weight to elements outside the domain on which it was defined.

The first correction keeps the original recurrence. It retains the unused degree allowance of the leading coefficient and proves

$$\Lambda(\beta_t) \le \tau + tw + e_t\sigma.$$

This bound is enough for all three places where Claim A.2 is used: removing poles, killing the high Hensel coefficients, and controlling the common numerator in the second resultant. The coarse numerical bounds in the published proximity-gap proof do not change.

The second correction clears denominators before taking weights. It constructs regular coefficients $\delta_t \in \mathcal{O}$ and proves

$$\Lambda(\delta_t) \le \tau + e_t\mu.$$

For $t > 0$, the identity $\delta_t = W^{t-1}\beta_t$ shows that the two estimates are the same calculation written with different numerators. The advantage of the second form is conceptual: every object to which a weight is assigned lies in the regular quotient.

The 2026 refinement requires one additional correction. The weighted degree of $R_i(x_0, Y, Z)$ does not control the coefficients that appear after shifting X. The full

(Y, Z)-weighted degree of R_i does, and these full degrees still sum within the degree of the global interpolation polynomial. Specialization-content roots fit inside the same pole allowance. The same global threshold therefore survives in both the separable and inseparable factorizations.

In every case, the selected algebraic root is unchanged. What changes is the regular numerator used to represent it and the proof of its weight bound. This is enough to restore the line argument, its parameterized-curve extension, and the full-factor bookkeeping used in the later refinement.

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